

EC 9130

Distribution Automation and Smart Grid

HIGH IMPEDANCE FAULT DETECTION

GROUP MEMBERS

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1. Introduction

Electrical power distribution systems play a critical role in delivering electricity safely and reliably to end users. However, these systems are frequently exposed to various types of faults due to environmental conditions, equipment failures, and external disturbances. Among these, High Impedance Faults (HIFs) represent one of the most challenging fault types to detect and manage. A HIF occurs when an energized conductor comes into contact with a high-resistance surface such as dry soil, asphalt, sand, or tree branches. Unlike conventional faults, HIFs do not produce large fault currents; instead, they generate low-magnitude, irregular, and non-linear current signals.

Traditional protection schemes, such as overcurrent relays, are primarily designed to detect faults based on high current levels. As a result, they often fail to identify HIFs because the fault current remains within the normal operating range of the system. This limitation creates a critical protection gap, as undetected HIFs can lead to serious hazards including fire outbreaks, electric shock risks to the public, and long-term damage to electrical infrastructure. Therefore, improving the detection of HIFs is essential for enhancing system safety, reliability, and overall power quality.

The detection of HIFs is particularly difficult due to their stochastic and time-varying nature. The fault characteristics depend on several factors such as the type of contact surface, environmental conditions, and system configuration. Consequently, conventional analytical and threshold-based methods are often insufficient for reliable detection. Advanced signal processing techniques, combined with intelligent algorithms, have emerged as promising solutions to address this challenge.

In this project, a three-phase distribution system model is developed using MATLAB/Simulink to simulate real-world operating conditions and fault scenarios. High impedance faults are modelled within this system to study their impact on voltage and current waveforms. To effectively capture the non-stationary characteristics of HIF signals, wavelet transform-based signal analysis is employed for feature extraction from phase current signals. Furthermore, a machine learning-based approach is considered to classify and accurately identify HIF events by distinguishing them from normal operating conditions and switching disturbances.

2. Motivation and Overview

Electrical power distribution systems play a vital role in delivering electricity from generation sources to end users in a safe and reliable manner. However, these systems are highly exposed to faults due to environmental conditions such as storms, lightning, falling trees, and equipment failures. Among the various types of faults, High Impedance Faults (HIFs) are considered one of the most critical and challenging problems in modern power systems. A HIF occurs when an energized conductor comes into contact with a high-resistance surface such as dry soil, asphalt roads, sand, or tree branches. In such conditions, the fault current is significantly limited due to the high resistance at the fault point.

Unlike conventional short-circuit faults, HIFs do not produce large fault currents. Instead, they generate low magnitude, irregular, asymmetric, and nonlinear current waveforms, often accompanied by arcing behavior. These characteristics make HIFs extremely difficult to detect using traditional protection systems such as overcurrent relays, which rely primarily on current magnitude thresholds. As a result, HIFs may remain undetected for extended periods, posing serious risks such as electric shock hazards to humans and animals, fire outbreaks, and long-term damage to electrical infrastructure.

In recent years, the increasing complexity of power systems and the integration of smart grid technologies have emphasized the need for intelligent fault detection mechanisms. Conventional signal processing techniques alone are not sufficient to handle the nonlinear and time-varying nature of HIF signals. Therefore, modern research focuses on combining advanced signal processing methods with machine learning techniques to improve detection accuracy.

This research proposes a hybrid approach that integrates Wavelet Transform (WT) for extracting meaningful features from current signals and Long Short-Term Memory (LSTM) neural networks for classification. Wavelet Transform is particularly effective in analyzing non-stationary signals, while LSTM networks are capable of capturing temporal dependencies in time-series data. By combining these techniques, the proposed system aims to accurately detect HIF conditions and enhance the reliability and safety of distribution networks.

3. Aim

The primary aim of this research is to develop an intelligent and robust High Impedance Fault detection system for power distribution networks by integrating advanced signal processing techniques with deep learning models. The proposed system aims to overcome the limitations of conventional protection methods by accurately identifying low-magnitude, nonlinear, and time-varying fault signals, thereby improving system safety, reliability, and operational efficiency.

4. Objectives

- The first objective is to develop a realistic simulation model of a three-phase electrical distribution system using MATLAB/Simulink. This model should be capable of representing normal operating conditions as well as various fault scenarios, particularly High Impedance Faults. Accurate modeling is essential to generate reliable data for analysis and testing.
- The second objective is to analyze the characteristics of High Impedance Fault signals, including their nonlinear behavior, low current magnitude, harmonic content, and randomness. Understanding these characteristics is crucial for designing an effective detection system.
- The third objective is to apply Wavelet Transform (WT) for feature extraction. Since HIF signals are non-stationary, wavelet transform is used to decompose the signals into different frequency components and extract meaningful features such as energy coefficients and statistical parameters.
- The fourth objective is to design and implement a Long Short-Term Memory (LSTM) based classification model. LSTM networks are chosen due to their ability to learn temporal patterns in sequential data, making them suitable for analyzing time-series electrical signals.
- Another objective is to train and evaluate the performance of the proposed model using simulated datasets. Performance metrics such as accuracy, precision, recall, and confusion matrix analysis are used to assess the effectiveness of the model.

5. Methodology

5.1. Overview of the Proposed System

The proposed High Impedance Fault (HIF) detection system is developed using a hybrid framework that integrates simulation-based data generation, signal processing, and deep learning techniques. The primary objective of this methodology is to accurately detect HIF conditions in a three-phase power distribution system and identify the faulted phase with high reliability.

Initially, a rule-based detection system is implemented using wavelet-derived features extracted from current signals. This baseline system is capable of detecting faults using threshold logic. However, due to the non-linear, non-stationary, and time-varying nature of HIF signals, the system is further enhanced by incorporating a Long Short-Term Memory (LSTM) neural network to enable intelligent classification and improved robustness.

The complete methodology consists of the following stages:

- Power system modelling and fault simulation
- Signal acquisition and preprocessing
- Sliding window-based segmentation
- Wavelet-based feature extraction
- Threshold-based detection and phase identification
- Dataset generation and labelling
- Sequence formation
- LSTM-based classification

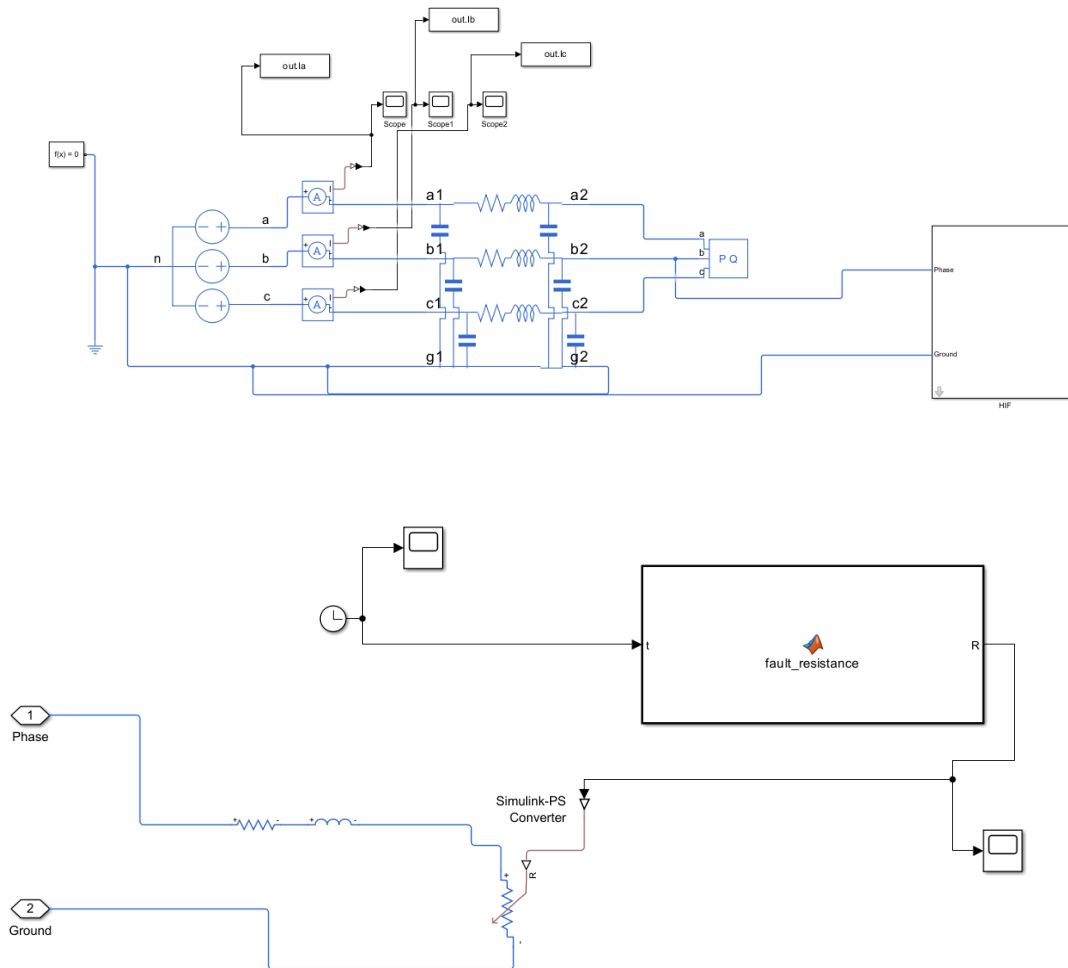
5.2. Power System Modelling and Fault Simulation

A detailed three-phase power distribution system is modelled using MATLAB/Simulink. The system includes an AC voltage source, transmission line, load, and measurement blocks to capture current signals.

High impedance faults are simulated by introducing a high-resistance path between the phase conductor and ground. This configuration mimics real-world HIF conditions, where the fault current is limited due to the high resistance of the fault path.

To generate a comprehensive dataset, multiple fault scenarios are simulated by varying the following parameters:

- Faulted phase (Phase A, Phase B, Phase C)
- Fault resistance (various high resistance values)
- Fault location along the feeder
- Fault initiation time (e.g., 0.02 seconds)
- This variability ensures that the dataset captures a wide range of operating conditions and improves the generalization capability of the detection system.



5.3. Signal Acquisition and Preprocessing

The three-phase current signals $I_a(t)$, $I_b(t)$, and $I_c(t)$ are extracted from the simulation output. These signals represent the instantaneous current values for each phase. Due to the variable time-step nature of the Simulink solver, the output signals are not uniformly sampled. Therefore, a resampling process is applied to convert the signals into a uniformly sampled time series.

$$t_{\text{uniform}} = 0:\frac{1}{F_s}:t_{\text{end}}$$

where F_s is the sampling frequency, selected as 10 kHz.

Linear interpolation is used to obtain uniformly sampled signals. This step ensures consistency in subsequent signal processing and feature extraction stages.

```
%% Get simulation data

Ia = out.Ia;
Ib = out.Ib.Data;
Ic = out.Ic.Data;
tout = out.tout;

%% 2 Resample
Fs_target = 10000;
t_uniform = 0:1/Fs_target:tout(end);

Ia_u = interp1(tout, Ia, t_uniform, 'linear');
Ib_u = interp1(tout, Ib, t_uniform, 'linear');
Ic_u = interp1(tout, Ic, t_uniform, 'linear');
```

5.4. Sliding Window Segmentation

To analyse the time-varying characteristics of HIF signals, a sliding window approach is adopted. The resampled signals are divided into overlapping windows, allowing localized analysis of signal features.

The window parameters are defined as:

Window length $T_w=20$ ms

Step size $T_s=10$ ms

This results in a 50% overlap between consecutive windows, which improves temporal resolution and enables early fault detection. Each window is processed independently to extract

```
%% 3 Sliding window
window_length = 0.02; % 20 ms
step = 0.01; % 10 ms
samples_per_window = round(window_length*Fs_target);
step_samples = round(step*Fs_target);

threshold = 0.01;
counter = 1;

for k = 1:step_samples:length(Ia_u)-samples_per_window
    idx = k:k+samples_per_window-1;

    HF_A(counter) = wavelet_index(Ia_u(idx), Fs_target);
    HF_B(counter) = wavelet_index(Ib_u(idx), Fs_target);
    HF_C(counter) = wavelet_index(Ic_u(idx), Fs_target);

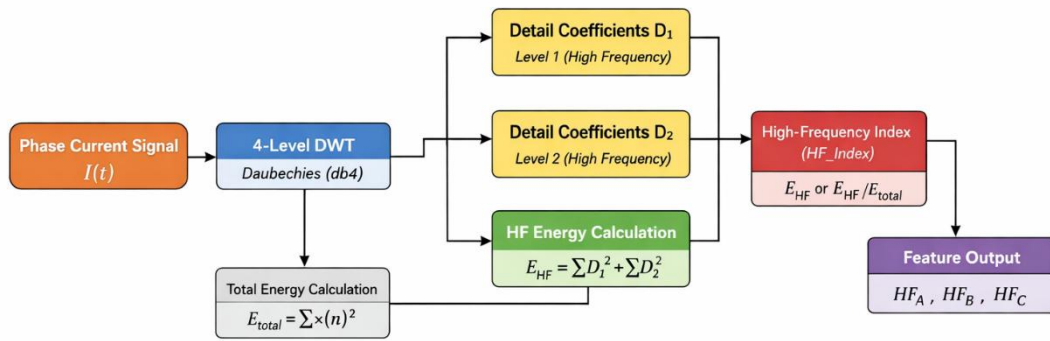
    time_axis(counter) = t_uniform(k);

    counter = counter + 1;
end
```

features. The use of overlapping windows ensures that transient events are not missed and improves detection reliability.

5.5. Wavelet-Based Feature Extraction

High Impedance Fault signals are inherently non-stationary and contain transient components due to arcing phenomena. Therefore, Discrete Wavelet Transform (DWT) is used for feature extraction.



Wavelet-Based High-Frequency Feature Extraction Process

A 4-level wavelet decomposition is performed using the Daubechies-4 (db4) mother wavelet. This wavelet is selected due to its excellent time-frequency localization properties and suitability for transient signal analysis.

The signal decomposition is expressed as:

$$x(t) = \sum_k a_{J,k} \phi_{J,k}(t) + \sum_{j=1}^J \sum_k d_{j,k} \psi_{j,k}(t)$$

The detail coefficients at the first two levels represent the high-frequency components:

D1=Detail coefficients at level 1

D2=Detail coefficients at level 2

The energy of these components is computed as:

$$E_{HF} = \sum_n D_1(n)^2 + \sum_n D_2(n)^2$$

The total energy of the signal is given by:

$$E_{total} = \sum_n x(n)^2$$

The High-Frequency Index is defined as:

$$HF_{index} = E_{HF}$$

or in normalized form:

$$HF_{index} = \frac{E_{HF}}{E_{total}}$$

This index effectively captures the presence of high-frequency disturbances associated with HIF conditions.

5.6. Phase-Wise Feature Computation

The feature extraction process is applied independently to each phase current. This results in three feature values corresponding to the three phases:

$$F = [HF_A, HF_B, HF_C]$$

This feature vector allows both detection of fault conditions and identification of the faulted

```
function HF_index = wavelet_index(signal, Fs)

[C,L] = wavedec(signal,4,'db4');
D1 = detcoef(C,L,1);
D2 = detcoef(C,L,2);

HF_energy = sum(D1.^2) + sum(D2.^2);
Total_energy = sum(signal.^2);

HF_index = HF_energy;

end
```

phase.

5.7. Removal of Initial Transients

To avoid false detections caused by system startup conditions, the initial portion of the signal is ignored. Specifically, the first 0.08 seconds of data are excluded from analysis.

This ensures that only steady-state conditions are considered for fault detection.

5.8. Threshold-Based Fault Detection

A rule-based detection mechanism is implemented using the extracted features. For each time

window, the maximum HF index among the three phases is compared with a predefined threshold:

$$\max(HF_A, HF_B, HF_C) > \text{Threshold}$$

If the condition is satisfied, a fault is detected, and the corresponding phase is identified.

5.9. Decision Confirmation Mechanism

To improve detection reliability, a confirmation mechanism is introduced. A fault is confirmed only if the same phase is detected over multiple consecutive windows.

$$P(i) = P(i + 1) = P(i + 2) \neq 0$$

This approach reduces false positives caused by noise or transient disturbances.

5.10. Detection Time Estimation

The fault detection time is estimated as the midpoint of the detection window:

$$t_{detect} = t_{window} + \frac{T_w}{2}$$

This provides a more accurate estimation of the fault occurrence time.

5.11. Dataset Generation and Labelling

The extracted features are used to generate a labelled dataset. Each sample consists of the feature vector and a corresponding label indicating the system condition.

Label 0: Normal condition

Label 1: High Impedance Fault

The dataset is used for training and evaluating the LSTM model.

5.12. Sequence Formation for LSTM

To capture temporal dependencies, feature vectors are arranged into sequences:

$$X_t = [F(t - n + 1), F(t - n + 2), \dots, F(t)]$$

This sequence-based representation enables the model to learn the evolution of fault characteristics over time.

5.13. LSTM-Based Fault Detection

A Long Short-Term Memory (LSTM) neural network is used for classification. LSTM is well suited for this application due to its ability to handle sequential data and capture long-term dependencies.

The LSTM network consists of:

- Input layer
- LSTM layer
- Fully connected layer
- Output layer

The model is trained using binary cross-entropy loss and optimized using the Adam optimizer.

The output is defined as:

$$y = \begin{cases} 1, & \text{HIF Detected} \\ 0, & \text{Normal Condition} \end{cases}$$

5.14. Performance Evaluation

The performance of the proposed system is evaluated using standard metrics such as accuracy, precision, recall, and F1-score. These metrics provide a comprehensive assessment of detection performance.

5.15. Advantages of the Proposed Method

- The proposed methodology offers several advantages:
- Accurate detection of low-current faults
- Ability to identify faulted phase
- Improved robustness using temporal learning
- Reduced false detections

5.16. Limitations and Future Work

- Despite its effectiveness, the system has some limitations:
- Dependence on simulation data
- Computational complexity of LSTM
- Need for parameter tuning
- Future improvements may include hybrid models, real-time implementation, and integration with smart grid systems.

6. Literature review

6.1. High Impedance Faults

High Impedance Faults (HIFs) are one of the most complex and dangerous fault conditions in modern power distribution systems. These faults occur when an energized conductor comes into contact with a high-resistance surface such as dry soil, asphalt, or tree branches, resulting in fault currents that are too low to be detected by conventional protection devices (A Simple Approach to Detect High Impedance Fault Using Morphological Gradient Edge Detector[1]).

Unlike low-impedance faults, HIFs do not produce significant current spikes, and their behaviour is highly nonlinear, asymmetric, and random due to arcing phenomena (Reliable High Impedance Fault Detection with Experimental Investigation in Distribution Systems[2]). This makes them particularly hazardous, as they can persist undetected and pose risks such as fire hazards and electric shock.

With the increasing complexity of power systems, including renewable integration and DC microgrids, the challenge of HIF detection has become even more significant. Traditional protection methods are no longer sufficient, leading to the development of advanced signal processing, machine learning, and hybrid approaches (High Resistance Fault Detection in DC Microgrid using Hilbert Huang Transform and Vector-Based Ensemble Optimized LSTM Networks[3]). This review analyses eight major research contributions, covering classical methods, signal processing techniques, artificial intelligence approaches, and modern deep learning-based solutions.

The existing literature on high impedance fault (HIF) detection shows a clear shift from conventional feature-based methods to more intelligent and adaptive approaches. One of the earlier methods was proposed by Sheng and Rovnyak [4], who used a decision tree with basic current features such as RMS current, harmonic magnitudes, and third-harmonic phase. The main strength of this method is its simplicity and low hardware requirement, since only current measurements with a relatively low sampling rate are needed. This makes it more practical than many later advanced methods. However, compared with recent intelligent approaches, its main limitation is that it relies on predefined features and was mainly validated using simulation data, so its generalization to real and changing field conditions is limited. Moreover, Sedighi et al. [5] moved beyond this simpler structure by combining wavelet transform with soft computing techniques such as GA–Bayes and PCA–NN. Compared with the decision tree method, this work used stronger feature extraction and better classification capability. Another advantage is

that it included both experimental and simulated data, which gives it more practical value than studies based only on simulations. At the same time, this method is more complex than the decision tree approach because it needs decomposition, feature reduction, and classification in multiple stages. So, while it improves detection capability, it also increases implementation complexity. Furthermore, Samantaray et al. [6] further improved the fault detection process by using S-transform and TT-transform with a probabilistic neural network (PNN). Compared with Sedighi et al. [5], this method offers stronger time–frequency analysis, which is important because HIF signals are highly non-stationary and difficult to distinguish from normal events. The authors also showed that HIFs could be identified within one cycle from fault inception, which is faster than many earlier approaches. However, compared with Sedighi et al. [5], this method was more dependent on simulated conditions and did not show the same level of practical experimental support. So, it improved signal representation and detection speed, but its real-world robustness remained less proven.

A similar wavelet-based direction was later studied by Silva et al. [7], but with an important difference: they used an evolving neural network instead of a fixed classifier. Compared with Sedighi et al. [5] and Samantaray et al. [6], this is a major improvement because those earlier methods used classifiers with fixed structures. Silva et al. [7] showed that fixed classifiers such as MLP, PNN, and SVM may perform well when the fault pattern remains similar, but their performance drops when the pattern changes over time. In contrast, the evolving neural network can adapt during operation. Therefore, compared with earlier wavelet-neural methods, this work is more suitable for real HIF conditions where the signal behavior is time-varying and non-stationary.

Moreover, Salehi et al. [1] proposed a different direction by introducing a morphological gradient edge detector (MGED). Compared with wavelet-based and neural-network-based methods, this approach is much simpler. It focuses on extracting edges or sudden signal changes from voltage signals and then forming a detection index. This reduces the dependence on deep learning models or complex transform selection. Compared with Silva et al. [7], the MGED approach may be easier to implement and computationally lighter. However, it is less adaptive than evolving learning methods and may not capture fault variability as effectively. Thus, its main advantage is simplicity, while its limitation is lower intelligence and flexibility when compared with adaptive machine learning approaches. Consequently, Satea et al. [2] proposed a more reliable hybrid method by combining wavelet analysis with third-harmonic information of zero-sequence current and voltage. Compared with Salehi et al. [1], this method is more elaborate because it does not rely only on one processed indicator. Instead, it adds another

physical feature to improve discrimination between HIFs and switching events. Compared with conventional wavelet-based methods, this gives better security against false operation. A strong advantage of this work over many previous studies is that it included experimental validation on an actual feeder, which makes its findings more convincing for practical application. Therefore, compared with both classical wavelet methods and simpler morphology-based methods, this study provides a better balance between detection sensitivity and practical reliability.

The most recent development can be seen in the 2025 study by Kumar et al. [3], which used Hilbert-Huang Transform (HHT) with vector-based ensemble optimized LSTM networks for high-resistance fault detection in DC microgrids. Compared with all the earlier AC distribution studies, this method represents a more advanced and modern framework. Unlike the decision tree, wavelet classifier, or morphology-based methods, the HHT-LSTM model is designed specifically to handle nonlinear and non-stationary signals while also learning long-term temporal patterns. This gives it a clear advantage in complex systems. However, compared with the older methods, it is also the most computationally demanding and model-heavy. So, while it offers higher intelligence and likely better accuracy, it may be more difficult to implement in practical protection hardware than simpler methods such as decision trees or MGED.

In addition, Hazarika and Roy [8] provided useful background through MATLAB/Simulink-based distribution system fault analysis. Although their work is not specifically focused on HIF detection, it helps explain the broader protection environment of distribution networks. Compared with the HIF-specific studies, this paper is more general and foundational. Its value lies in giving a clear understanding of fault types, system behavior, and the need for fast fault clearance, which supports the discussion of why accurate HIF detection is necessary.

Overall, these studies can be compared in three main ways. First, in terms of complexity, Sheng and Rovnyak [4] and Salehi et al. [1] proposed simpler methods, while Samantaray et al. [6], Silva et al. [7], Satea et al. [2], and Kumar et al. [3] developed more advanced and computationally heavier frameworks. Second, in terms of adaptability, fixed classifiers such as decision trees, Bayes, standard neural networks, and PNN are weaker than evolving neural networks and LSTM-based approaches. Third, in terms of practical reliability, studies with experimental validation, such as Sedighi et al. [5] and Satea et al. [2], are stronger than purely simulation-based works. Therefore, the literature suggests that no single method is perfect: simpler methods are easier to implement, but adaptive hybrid methods are generally more robust and accurate for real HIF detection.

6.2. Characteristics and Challenges of HIF Detection

HIF signals exhibit several distinctive characteristics that complicate their detection. These include low current magnitude, asymmetrical waveform behaviour, intermittent arcing, and the presence of harmonics and high-frequency components (Reliable High Impedance Fault Detection with Experimental Investigation in Distribution Systems[2]).

The nonlinear relationship between voltage and current in HIF conditions leads to irregular waveform patterns, making it difficult to model these faults accurately (HIF Detection Using Wavelet Transform and Evolving Neural Network). Additionally, HIF signals are non-stationary, meaning their frequency content changes over time, requiring advanced analysis techniques beyond traditional Fourier methods.

Another major challenge is the similarity between HIF signals and normal system disturbances such as capacitor switching, transformer inrush currents, and load variations (Decision Tree-Based Methodology for High Impedance Fault Detection[4]). These similarities often lead to false detections if simple threshold-based methods are used.

Furthermore, environmental factors such as surface type, weather conditions, and grounding characteristics significantly affect HIF behaviour, adding another layer of complexity (Soft Computing Applications in High Impedance Fault Detection[5]). Therefore, robust detection systems must be adaptive and capable of handling uncertainty and noise.

6.3. Time-Domain Detection Methods

One of the simplest approaches to HIF detection is based on time-domain signal analysis. The paper A Simple Approach to Detect High Impedance Fault Using Morphological Gradient Edge Detector[1] introduces a technique based on mathematical morphology.

This method applies morphological operations such as dilation and erosion to extract edges from voltage signals. The Morphological Gradient Edge Detector (MGED) highlights abrupt changes associated with fault conditions. Based on the number and pattern of detected edges, a High Impedance Fault Detection Index (HIFDI) is calculated to classify events.

This approach is advantageous due to its low computational requirements and suitability for real-time implementation (A Simple Approach to Detect High Impedance Fault Using Morphological Gradient Edge Detector[1]). It can effectively distinguish between HIFs and other disturbances such as switching events and CT saturation.

However, time-domain methods have limitations. They are highly sensitive to noise and may fail in complex scenarios where fault signatures are subtle or masked by other disturbances (Reliable High Impedance Fault Detection with Experimental Investigation in Distribution

Systems[2]). Additionally, they lack frequency-domain information, which is essential for analyzing non-stationary signals.

6.4. Time–Frequency Analysis Methods

To address the limitations of time-domain techniques, researchers have introduced time–frequency analysis methods. The paper HIF Detection Using Time-Frequency Transform and Probabilistic Neural Network[6] Employs S-transform and TT-transform to analyse signals in both time and frequency domains.

These transforms provide localized frequency information, enabling the detection of transient and harmonic components associated with HIFs. Features such as energy distribution and statistical parameters are extracted and used as inputs to a Probabilistic Neural Network (PNN) (HIF Detection Using Time–Frequency Transform and Probabilistic Neural Network[6]).

The integration of signal processing and machine learning improves detection accuracy and robustness. The PNN classifier can effectively distinguish between fault and non-fault conditions even under nonlinear loading scenarios.

However, this approach has certain drawbacks. The computational complexity of time–frequency transforms can be high, making real-time implementation challenging (HIF Detection Using Time-Frequency Transform and Probabilistic Neural Network[6]). Additionally, the performance depends heavily on the quality of feature extraction and the availability of sufficient training data.

6.5. Wavelet Transform-Based Methods

Wavelet Transform (WT) has emerged as one of the most effective tools for analysing non-stationary signals in HIF detection.

6.6. Discrete Wavelet Transform (DWT)

The paper HIF Detection Using Wavelet Transform and Evolving Neural Network[7] uses Discrete Wavelet Transform (DWT) to decompose current signals into multiple frequency bands. This allows the extraction of high-frequency components associated with arcing faults. DWT provides multi-resolution analysis, enabling the detection of both transient and steady-state features. These features are then used for classification using neural networks (HIF Detection Using Wavelet Transform and Evolving Neural Network[7]).

6.7. Wavelet + Evolving Neural Networks

The evolving neural network used in this study adapts its structure dynamically based on incoming data. This allows the system to handle non-stationary signals and changing system conditions effectively.

Compared to traditional neural networks, evolving neural networks offer better adaptability and improved performance in real-world scenarios (HIF Detection Using Wavelet Transform and Evolving Neural Network[7]).

6.8. Limitations

Despite their advantages, wavelet-based methods have limitations. The selection of the mother wavelet and decomposition level significantly affects performance. Additionally, wavelet transforms can be sensitive to noise and may require preprocessing techniques (Soft Computing Applications in High Impedance Fault Detection[5]).

6.9. Soft Computing and Feature Optimization

The paper Soft Computing Applications in High Impedance Fault Detection[5] introduces a hybrid approach combining wavelet transform with optimization techniques such as Genetic Algorithms (GA) and Principal Component Analysis (PCA).

GA is used to select the most relevant features, while PCA reduces dimensionality by eliminating redundant information. The optimized features are then used for classification using neural networks or Bayesian classifiers.

This approach improves detection accuracy and reduces computational complexity by focusing on the most informative features (Soft Computing Applications in High Impedance Fault Detection[5]). It also enhances robustness against noise and system variations.

However, the use of optimization algorithms introduces additional computational overhead and requires careful parameter tuning.

6.10. Simulation-Based Fault Analysis

Simulation plays a crucial role in understanding fault behavior and generating datasets for training machine learning models.

The paper Distribution System Fault Analysis Using MATLAB/Simulink[8] presents a detailed simulation of various fault conditions, including line-to-ground, line-to-line, and three-phase faults. The study analyzes voltage and current waveforms to identify distinguishing features of different faults.

The results show that HIFs produce unique waveform distortions and harmonic patterns compared to conventional faults (Distribution System Fault Analysis Using MATLAB/Simulink[8]). These insights are essential for feature extraction and model training. Although this paper does not propose a detection algorithm, it provides a strong foundation for developing and validating HIF detection methods.

6.11. Hybrid Signal Processing Techniques

Hybrid approaches combine multiple signal processing methods to improve detection reliability.

The paper Reliable High Impedance Fault Detection with Experimental Investigation in Distribution Systems[2] proposes a method combining DWT and DFT. DWT is used to capture transient features, while DFT is used to analyze harmonic components.

The study focuses on the third harmonic component of zero-sequence current and voltage, which is a key indicator of HIF conditions. By combining these features, the method achieves high detection accuracy and effectively distinguishes between faults and switching events (Reliable High Impedance Fault Detection with Experimental Investigation in Distribution Systems[2]).

A major strength of this work is its experimental validation using real-world data, demonstrating practical applicability.

6.12. Classical Machine Learning Approaches

Machine learning techniques provide a data-driven approach to HIF detection.

The paper Decision Tree-Based Methodology for High Impedance Fault Detection[4] uses decision trees for classification. Features such as RMS current and harmonic components are extracted using FFT.

Decision trees offer several advantages, including simplicity, interpretability, and low computational cost. They can be easily implemented in protective relays (Decision Tree-Based Methodology for High Impedance Fault Detection[4]).

However, their performance depends heavily on feature selection and may not generalize well to unseen conditions.

6.13. Deep Learning and Advanced Hybrid Methods

The most recent advancements in HIF detection involve deep learning and hybrid approaches.

6.13.1. HHT + LSTM Approach

The paper High Resistance Fault Detection in DC Microgrid using Hilbert Huang Transform and Vector-Based Ensemble Optimized LSTM Networks[3] introduces a hybrid method combining Hilbert–Huang Transform (HHT) and Long Short-Term Memory (LSTM) networks.

HHT decomposes signals into intrinsic mode functions (IMFs), capturing non-linear and non-stationary behaviour. Features such as instantaneous amplitude and frequency are extracted and used as inputs to the LSTM model.

LSTM networks are capable of learning temporal dependencies, making them suitable for time-series analysis. The combination of HHT and LSTM provides a powerful tool for HIF detection in DC microgrids (High Resistance Fault Detection in DC Microgrid using Hilbert Huang Transform and Vector-Based Ensemble Optimized LSTM Networks[3]).

6.13.2. Performance Analysis

The proposed method achieves detection accuracy between 92% and 98%, outperforming traditional methods such as DWT and FFT-based approaches (High Resistance Fault Detection in DC Microgrid using Hilbert Huang Transform and Vector-Based Ensemble Optimized LSTM Networks[3]).

It also demonstrates robustness under varying fault conditions and noise levels, making it suitable for real-world applications.

6.13.3. Limitations

Despite its high performance, the method has limitations. It requires significant computational resources and large training datasets. Additionally, model performance depends on hyperparameter tuning.

6.13.4. Comparative Analysis

The reviewed methods can be compared as follows:

- Time-domain methods are simple but lack accuracy
- Time–frequency methods provide better feature extraction but are computationally intensive
- Wavelet-based methods offer multi-resolution analysis but depend on parameter selection
- Machine learning methods improve classification but require feature engineering
- Hybrid methods provide higher accuracy and reliability
- Deep learning methods offer the best performance but at the cost of complexity

6.13.5. Research Gaps

Several research gaps remain:

- Lack of real-time implementations
- Limited datasets for training
- Difficulty in handling noise and uncertainty
- Need for lightweight models for embedded systems
- Limited studies on DC microgrids

6.13.6. Future Directions

Future research should focus on:

- Hybrid models combining wavelet, HHT, and deep learning
- Real-time embedded implementations
- Improved datasets using simulation and field data
- Integration with smart grid technologies

7. Results and observations

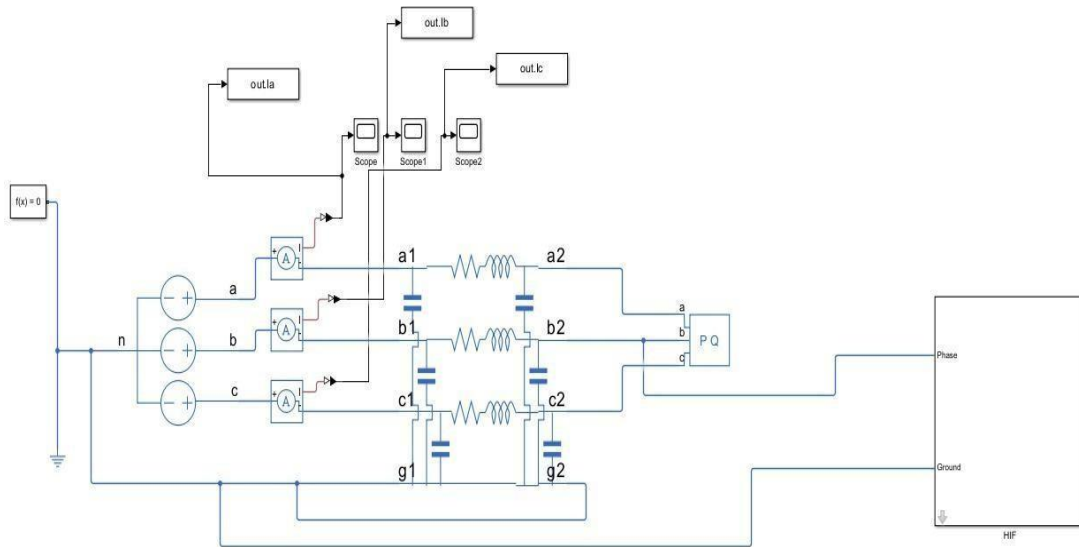


Fig 1: Proposed three-phase electrical system

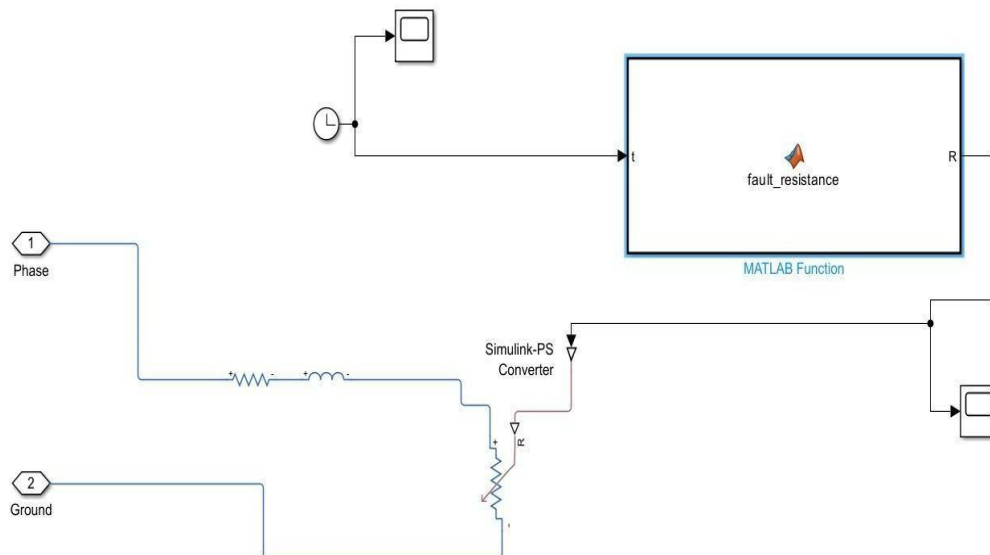
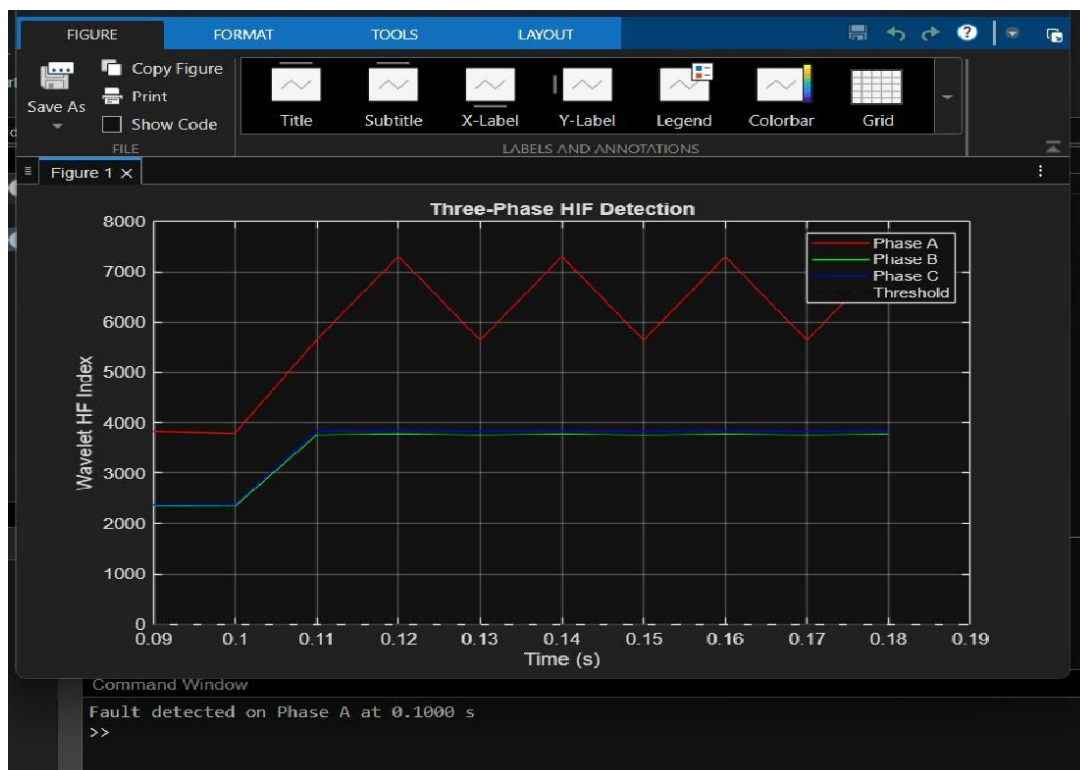


Fig 2: Proposed algorithm for identify HIFs

Results



Using LSTM

```
LSTM.py 6 X Release Notes: 1.111.0
C: > Users > ADMIN > Downloads > Group-20 (1) > Group-20 > LSTM.py > ...

102 # --- 4. TRAINING & EVALUATION ---
103
104 print("\nStarting Training.....")
105 history = model.fit(x_train, y_train,
106                     epochs=30,
107                     batch_size=32,
108                     validation_data=(x_test, y_test),
109                     verbose=1)
110
111 # Final Diagnostics
112 y_pred = (model.predict(x_test) > 0.5).astype(int)
113 print("\n--- Confusion Matrix ---")
114 print(confusion_matrix(y_test, y_pred))
115 print("\n--- Final Report ---")
116 print(classification_report(y_test, y_pred))
```

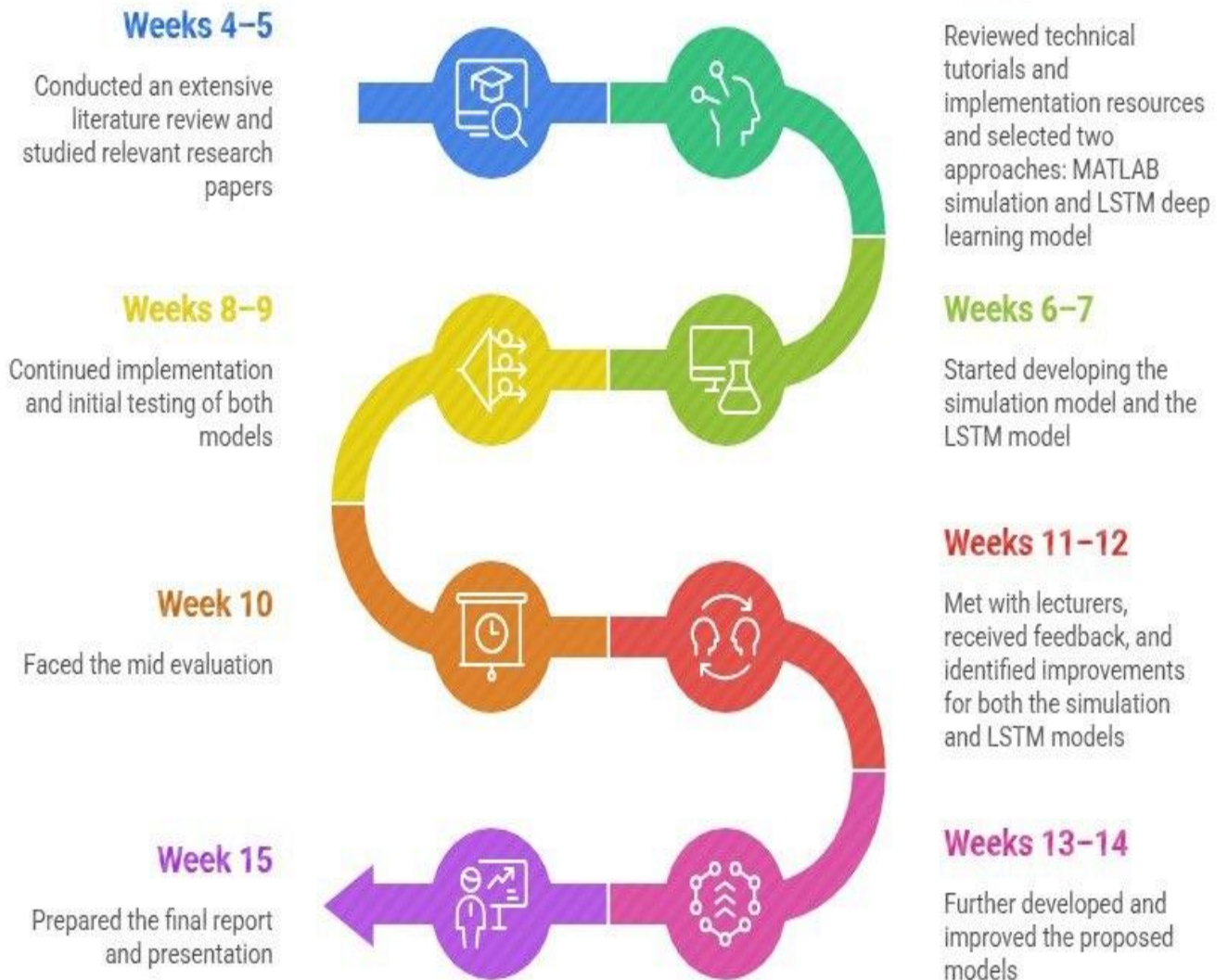
```
LSTM.py 6 X Release Notes: 1.111.0
C: > Users > ADMIN > Downloads > Group-20 (1) > Group-20 > LSTM.py > ...

69 # --- 2. PRE-PROCESSING ---
70
71 print("Generating Challenging 50Hz Dataset...")
72 x_raw, y_raw = generate_challenging_50hz()
73
74 # Reshape for LSTM (Samples, Time Steps, 1)
75 x = x_raw.reshape((x_raw.shape[0], x_raw.shape[1], 1))
76
77 # Z-Score Normalization (Standardization)
78 # This is much harder for the model than simple 0-1 scaling
79 x = (x - np.mean(x, axis=1, keepdims=True)) / np.std(x, axis=1, keepdims=True)
80
81 x_train, x_test, y_train, y_test = train_test_split(x, y_raw, test_size=0.2, shuffle=True)
82
83 # --- 3. THE LSTM ARCHITECTURE ---
84
85 model = Sequential([
86     # Reduced capacity to prevent "overfitting" on synthetic math
87     LSTM(32, input_shape=(x.shape[1], 1), return_sequences=True),
88     BatchNormalization(),
89     Dropout(0.4),
90
91     LSTM(16),
92     Dropout(0.4),
93
94     Dense(16, activation='relu'),
95     Dense(1, activation='sigmoid')
96 ])
97
98 # Using a slower learning rate to force the model to look for deeper features
99 opt = tf.keras.optimizers.Adam(learning_rate=0.0005)
100 model.compile(optimizer=opt, loss='binary_crossentropy', metrics=['accuracy'])
101
102 # --- 4. TRAINING & EVALUATION ---
103
```

```
LSTM.py 6 x Release Notes: 1.111.0
C: > Users > ADMIN > Downloads > Group-20 (1) > Group-20 > LSTM.py > ...

1  # -*- coding: utf-8 -*-
2  """Untitled5.ipynb
3
4  Automatically generated by Colab.
5
6  Original file is located at
7  https://colab.research.google.com/drive/1_pgk4hE-cXYf4_l4rNIQ4kEljwN7y9Yr
8  """
9
10 import numpy as np
11 import tensorflow as tf
12 from tensorflow.keras.models import Sequential
13 from tensorflow.keras.layers import LSTM, Dense, Dropout, BatchNormalization
14 from sklearn.model_selection import train_test_split
15 from sklearn.metrics import confusion_matrix, classification_report
16
17 # --- 1. THE "REAL-WORLD" DATA GENERATOR ---
18
19 def generate_challenging_50hz(n_samples=1200, fs=4000, duration=0.2):
20     t = np.linspace(0, duration, int(fs * duration))
21     X, y = [], []
22
23     for _ in range(n_samples):
24         is_fault = np.random.choice([0, 1])
25
26         # A. Frequency Jitter: The grid varies between 49.9 and 50.1 Hz
27         actual_freq = np.random.uniform(49.9, 50.1)
28         i_peak = np.random.uniform(10, 50)
29
30         # B. Base Signal with Harmonics (Applied to BOTH classes)
31         # 3rd and 5th harmonics are common in ALL modern power systems
32         signal = i_peak * np.sin(2 * np.pi * actual_freq * t)
33         signal += (i_peak * np.random.uniform(0.02, 0.12)) * np.sin(2 * np.pi * (3 * actual_freq) * t)
34         signal += (i_peak * np.random.uniform(0.01, 0.05)) * np.sin(2 * np.pi * (5 * actual_freq) * t)
```

8. Timeline



9. Conclusion

The High Impedance Fault (HIF) detection project represents an essential step in improving the reliability and safety of electrical power distribution systems, especially considering the limitations of traditional detection methods. HIFs occur when a live conductor contacts a high-resistance surface, such as dry soil, asphalt roads, or tree branches, which results in a low current that standard overcurrent relays cannot detect. These undetected faults pose serious risks, including fire hazards, electrical shocks, and damage to equipment, thus highlighting the need for an intelligent, reliable HIF detection system.

The project's primary objective is to design and develop a three-phase distribution system capable of simulating high impedance faults and integrating an intelligent detection algorithm for precise identification of these faults. This methodology combines Wavelet Transform (WT) for feature extraction from the phase currents and Machine Learning (ML) for the classification and detection of fault conditions. The integration of WT with machine learning techniques, particularly Long Short-Term Memory (LSTM) networks, addresses the critical gap identified in previous studies, where methods failed to distinguish high impedance faults from normal switching events.

The literature review has provided a deep dive into current methodologies for fault detection, such as the use of wavelet analysis and decision tree-based approaches. However, these methods have limitations, such as their reliance on simulated data and an inability to effectively differentiate between high impedance faults and other disturbances. Furthermore, many of these systems have limited real-time application, highlighting the importance of further research into hybrid techniques that combine signal processing and machine learning for improved reliability.

One of the significant challenges identified in the literature was the inability of conventional fault detection techniques to distinguish between HIFs and normal system operations, such as switching events or transient conditions. Many studies focused on harmonics analysis or signal features but failed to provide a comprehensive solution that combines these features with machine learning for accurate and reliable fault detection. This is where our approach aims to make a notable impact by combining Wavelet Transform with LSTM networks, leveraging the strengths of both techniques to create a hybrid system capable of identifying high impedance faults more accurately.

The research gap we aim to address includes improving the accuracy of fault detection, real-time performance, and the ability to distinguish faults from normal conditions. The current progress in the project includes successful simulation of a three-phase distribution system using MATLAB/Simulink and HIF modeling, along with the initial development of an algorithm designed to identify these faults. Preliminary results from the simulations show promise, with the algorithm effectively detecting faults under various conditions. Further work will focus on optimizing the model's sampling frequency, parameter tuning, and refining the ML-based approach to enhance its precision.

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